B.Sc. in SWE, 1st Semester

October 23, 2024 **(Morning)**

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Mid-Semester Examination
Course No.: Phy 4143
Course Title: Physics II

Winter Semester, A. Y. 2023-2024
Time: 2 Hours
Full Marks: 120

There are 4 **(four)** questions. Answer all 4 **(four)** questions. The symbols have their usual meanings. Programmable calculators are not allowed. Marks of each question and corresponding COs and POs are written in the brackets.

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1. a) Explain the difference between direct current (DC) and alternating current (AC). Define active and passive elements in electrical circuits and provide two examples for each case. **6 (CO1, PO1)**

b) Find the power absorbed by each of the elements in Fig. 1(b). **10 (CO2, PO2)**

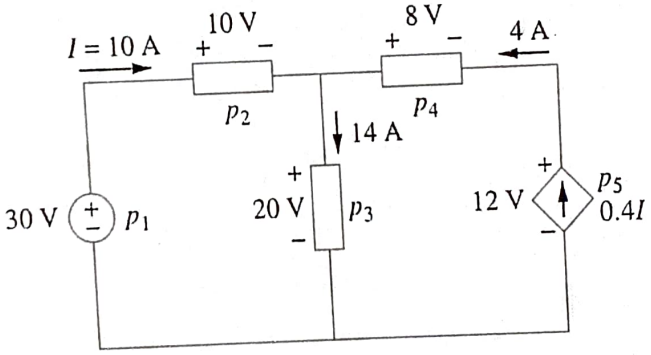


Fig. 1(b)

c) Determine the equivalent resistance R_{ab} for the circuit in Fig. 1(c) and use it to find current i. **14 (CO2, PO2)**

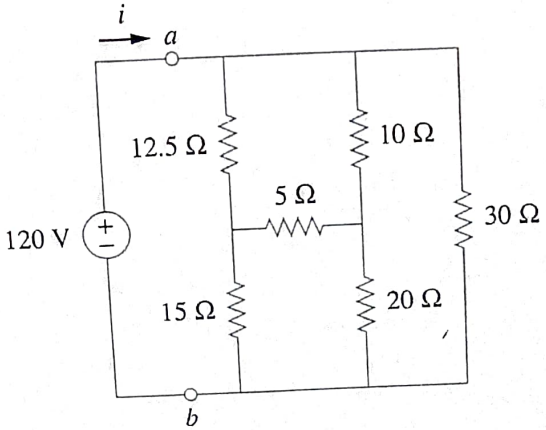


Fig. 1(c)

2. a) Find v_0 in the circuit in Fig. 2(b) and the power absorbed by the dependent source. (For solving the circuit, any method can be used) **10 (CO2, PO2)**

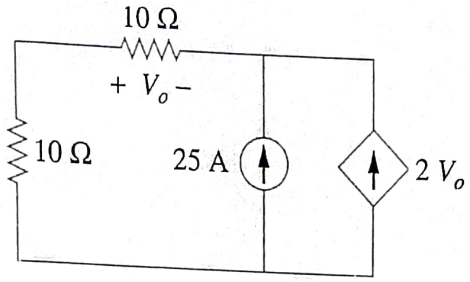


Fig. 2(a)

b) Define node, branch and loop in the context of electrical circuits. If the circuit shown in Fig. 2(b) has b branches, n nodes, and l independent loops, then determine the values of b, n, and l. Verify that, these obtained values satisfy the fundamental theorem of network topology: $b = l + n - 1$. **8 (CO2, PO2)**

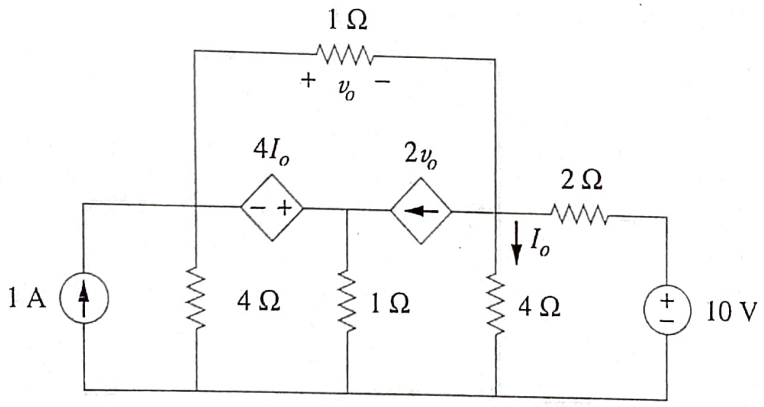


Fig. 2(b)

c) Determine v_0 in the circuit of Fig. 2(c) using the superposition principle. **12 (CO2, PO2)**

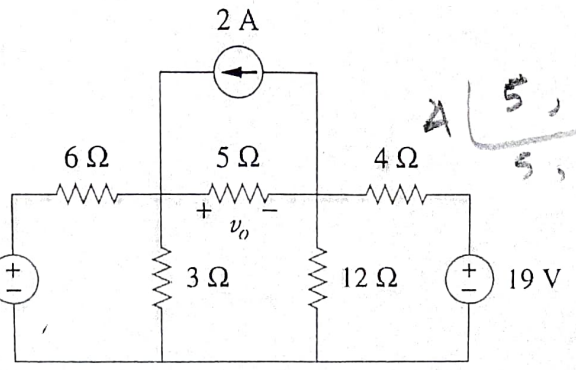


Fig. 2(c)

3. a) For the circuit in Fig. 3(a), determine v_1 and v_2 using nodal analysis. **8 (CO2, PO2)**

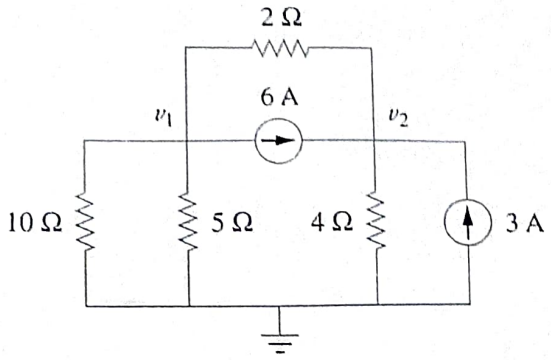


Fig. 3(a)

b) Determine the value of i for the circuit in Fig. 3(b) using mesh analysis. **10 (CO2, PO2)**

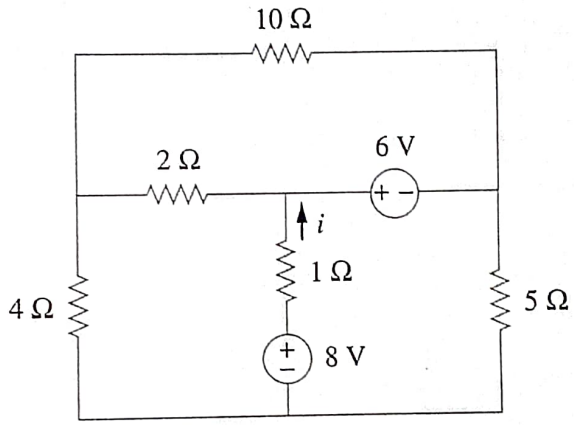


Fig. 3(b)

c) Find i_o in the circuit of Fig. 3(c) using superposition principle. **12 (CO2, PO2)**

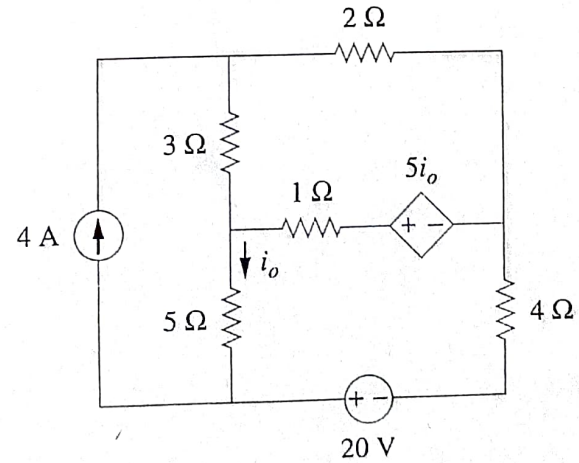


Fig. 3(c)

4. a) Find v_1, v_2, and v_3 in the circuit of Fig. 4(a) using nodal analysis. **12 (CO2, PO2)**

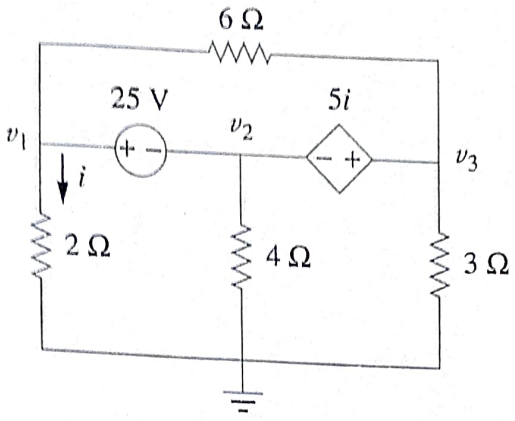


Fig. 4(a)

b) Find i_1, i_2, and i_3 in the circuit of Fig. 4(b) using mesh analysis. **12 (CO2, PO2)**

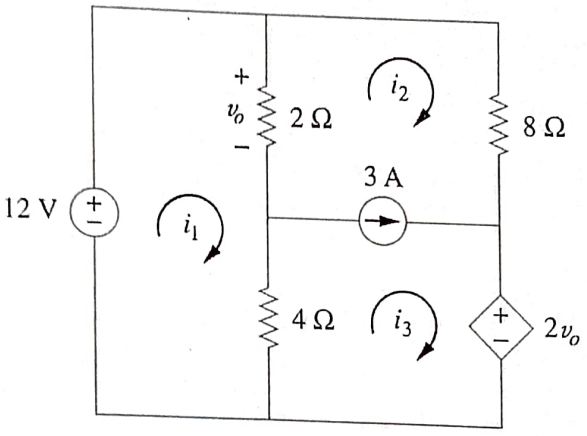


Fig. 4(b)

c) For the circuit in Fig. 4(c), find the equivalent resistance at terminals a-b. **6 (CO2, PO2)**

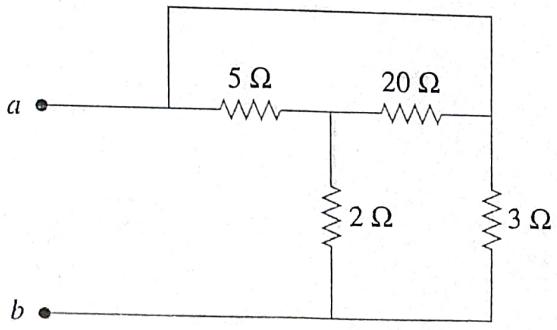


Fig. 4(c)